



Code Editor

Visual Editor



Recompile



S...



WRITEFULL



SmartPDF Chatbot.tex

```
1 \documentclass[12pt]{ar
2
3 \usepackage[a4paper,margin=1i
n]{geometry}
4 \usepackage{graphicx}
5 \usepackage{amsmath}
6 \usepackage{hyperref}
7 \usepackage{enumitem}
8 \usepackage{tikz}
9 \usepackage{float}
10 \usepackage[T1]{fontenc}
11 \usepackage[utf8]{inputenc}
12
13 \usetikzlibrary{shapes.geomet
ric, arrows.meta,
positioning}
14
15 \title{\textbf{ SmartPDF
Chatbot}}
16 \author{Kartik Tayal}
17 \date{}
18
19 \begin{document}
20
21 \maketitle
22
23 \section{Introduction}
24
25 This project implements a
Retrieval-Augmented
Generation (RAG) based AI
assistant capable of
answering questions from PDF
documents using semantic
retrieval and Large Language
Models (LLMs).
26
27 The system combines document
processing, embedding
generation, vector
similarity search, and
response generation to
provide context-aware
answers grounded in uploaded
```

SmartPD

Karti

1 Introduction

This project implements a Retrieval-Augmented Generation (RAG) based AI assistant capable of answering questions from PDF documents using semantic retrieval and Large Language Models (LLMs).

The system combines document processing, embedding generation, vector similarity search, and response generation to provide context-aware answers grounded in uploaded documents.

2 Project Objective

The objective of this project is to build an intelligent system capable of:

- Reading and processing PDF documents
- Understanding semantic meaning from text
- Retrieving contextually relevant information
- Generating grounded and accurate responses

3 Technology Stack

- Python
- LangChain
- ChromaDB
- HuggingFace Embeddings
- Google Gemini API
- PyPDF

4 System Workflow

The complete RAG workflow used in this project is as follows:

PDF → Text Extraction → Chunking → Embedding

5 RAG Pipeline Flowchart

File outline

Intr...

Proj...

Tech...

Syst...

RAG ...

Impl...

P...

T...

E...

V...

S...

R...

Chal...